



Course Title	Parallel and Distributed Computing
Instructor	Yousra Rehman
Assignment Number	1
Section	A
Program	BSCS
Student Name / Roll no	Mohammad Ibrahim / F23-2353

Assignment Release Date:	29/04/2026
Submission Date/Time:	30 /April /2026 12:00 (noon)

Assignment Questions

Question 1.

(5 marks)

Define Synchronous Communication and Asynchronous Communication in Parallel and Distributed Computing. Explain their work with suitable examples.

Answer 1:

Synchronous communication is a fundamental concept in parallel computing where the sender and the receiver are tightly coupled in terms of timing. When a task initiates a synchronous send operation, it enters a "blocked" state, meaning it cannot proceed with any further computation or instructions until it receives confirmation or acknowledgment that the message has been successfully received. This mechanism acts as a precise handshake, ensuring that the system works together properly and maintains strict data consistency. A clear example of this is a blocking MPI (Message Passing Interface) call, where a processor pauses its work until the destination node confirms the data transfer is complete, much like a real-time telephone conversation where both parties must be present and engaged simultaneously to exchange information.

Question 2.

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